Model Variants

The Go2 series is available in three distinct models — **AIR**, **PRO**, and **EDU**. Below is a comparative specification table for each model:

Model	AIR	PRO	EDU
Mechanical Specifications			
Dimension (Standing)	70cm x 31cm x 40cm	70cm x 31cm x 40cm	70cm x 31cm x 40cm
Dimension (Crouching)	76cm x 31cm x 20cm	76cm x 31cm x 20cm	76cm x 31cm x 20cm
Weight (with battery)	≈15kg	≈15kg	≈15kg
Material	Aluminium alloy + High-strength engineering plastic	Aluminium alloy + High-strength engineering plastic	Aluminium alloy + High-strength engineering plastic
Electronic Parameters			
Voltage	28V–33.6V	28V–33.6V	28V–33.6V
Max Working Power	≈3000W	≈3000W	≈3000W
Performance Parameters			
Payload	≈7kg (Max 10kg)	≈8kg (Max 10kg)	≈8kg (Max 12kg)
Speed	0–2.5 m/s (Max 5m/s)	0–3.5 m/s (Max 5m/s)	0–3.7 m/s (Max 5m/s)
Max Climb Drop Height	≈15cm	≈16cm	≈16cm
Max Climb Angle	30°	40°	40°
Computing Power	—	8-core High-perf CPU	8-core High-perf CPU

Model	AIR	PRO	EDU
Knee Joint Parameters			
Max Joint Torque	—	≈45N·m	≈45N·m
Knee Motors	12 sets	12 sets	12 sets
Range of Motion	Body: -48° ~ 48° Thigh: -200° ~ 90° Shank: -156° ~ -48°	Body: -48° ~ 48° Thigh: -200° ~ 90° Shank: -156° ~ -48°	Body: -48° ~ 48° Thigh: -200° ~ 90° Shank: -156° ~ -48°
Intra-joint Circuit	√	√	√
Heat Pipe Cooler	√	√	√
Sensors			
3D LiDAR (Wide-angle)	√	√	√
Wireless Position Module	—	√	√
HD Wide-angle Camera	√	√	√
Foot-end Force Sensor	—	—	√
Features			
Basic Actions	√	√	√
Auto-scaling Strap	—	√	—
Intelligent OTA Upgrade	√	√	√
RTT 2.0 Pictures	√	√	√
Graphical Programming	√	√	√
Front Lamp (3W)	√	√	√
Wi-Fi 6 Dual Band	√	√	√
Bluetooth 5.2/4.2/2.1	√	√	√
4G Module	—	√	√

Model	AIR	PRO	EDU
Voice Function	—	√	√
ISS 2.0 Side-follow	—	√	√
Obstacle Detection	√	√	√
Charging Pile Support	—	√	√
Secondary Development	—	—	√
Accessories			
Manual Controller	Optional	Optional	√
High Computing Module	—	—	NVIDIA Jetson Orin (Optional,40–100Tops)
Battery	8000mAh (x1)	8000mAh (x1)	15000mAh (x1)
Battery Life	1–2h	1–2h	2–4h
Charger	33.6V 3.5A	33.6V 3.5A	33.6V 9A (Fast)
Warranty	6 months	12 months	12 months

Note

1. Max joint torque refers to the largest joint motor. Actual torque varies by motor.
2. Specifications may vary in different scenarios/configurations. Refer to actual product.
3. Voice functions include offline commands, intercom, and media playback.
4. For development, refer to the **Secondary Development Manual**.
5. Warranty terms: See **Product Warranty Brochure**.
6. Non-mainland versions of 4G support 43 European countries and select Asian regions.

Tip

- Some functions require manual operation or secondary development due to hardware/software limits.

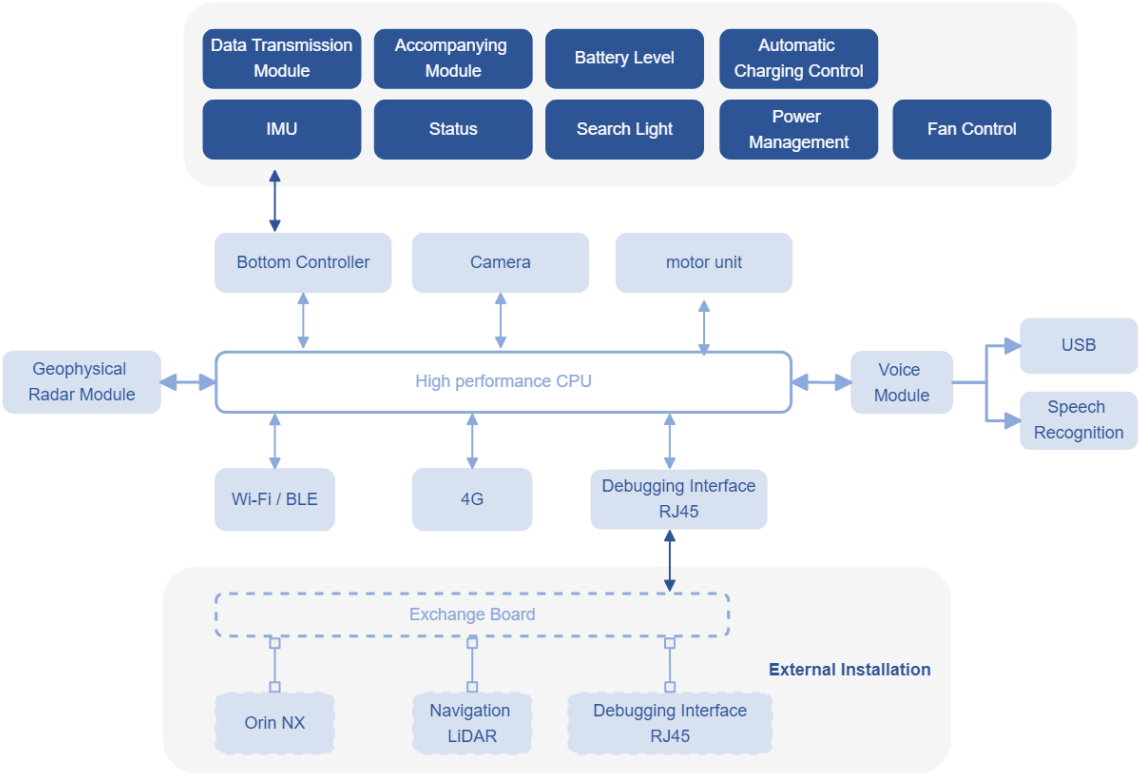
- Go2 is a **civilian-use robot**. Avoid unsafe modifications or hazardous use.
- Visit the **Unitree Robotics Website** for latest terms, updates, and compliance guidelines.

Accessories Overview

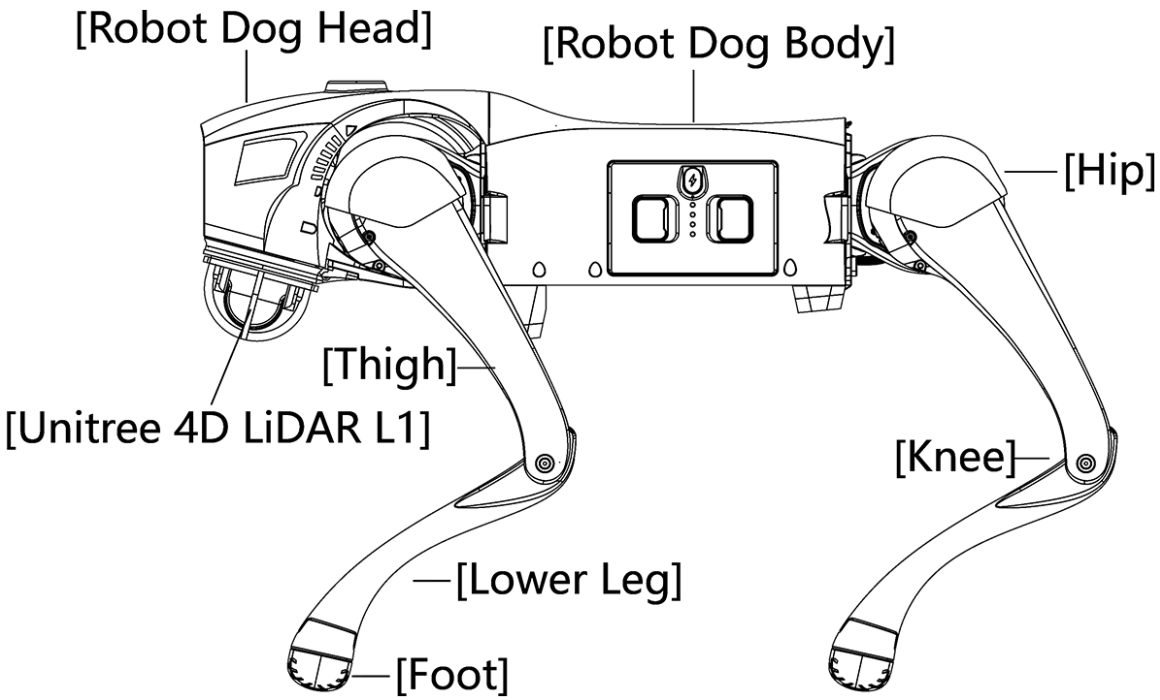
Accessory	Description	Notes	Availability	Picture
Manual Controller	Remote handheld controller	Optional for AIR & PRO; Standard in EDU	EDU	
Standard Battery	8000mAh / 236.8Wh	≈2.5kg	Default on AIR & PRO	
Long Endurance Battery	15000mAh / 432Wh	≈2.5kg	Default on EDU	
Charger (Standard)	33.6V 3.5A	Slow charge ~2h	AIR & PRO	
Fast Charger	33.6V 9A	Fast charge ~1h15	EDU + optional upgrade	
Foot End Sensor	Mounted on foot ends	Provides advanced force detection	EDU only	
Charging Pile	Dock-type charging pile	Supports auto-charging	PRO & EDU	

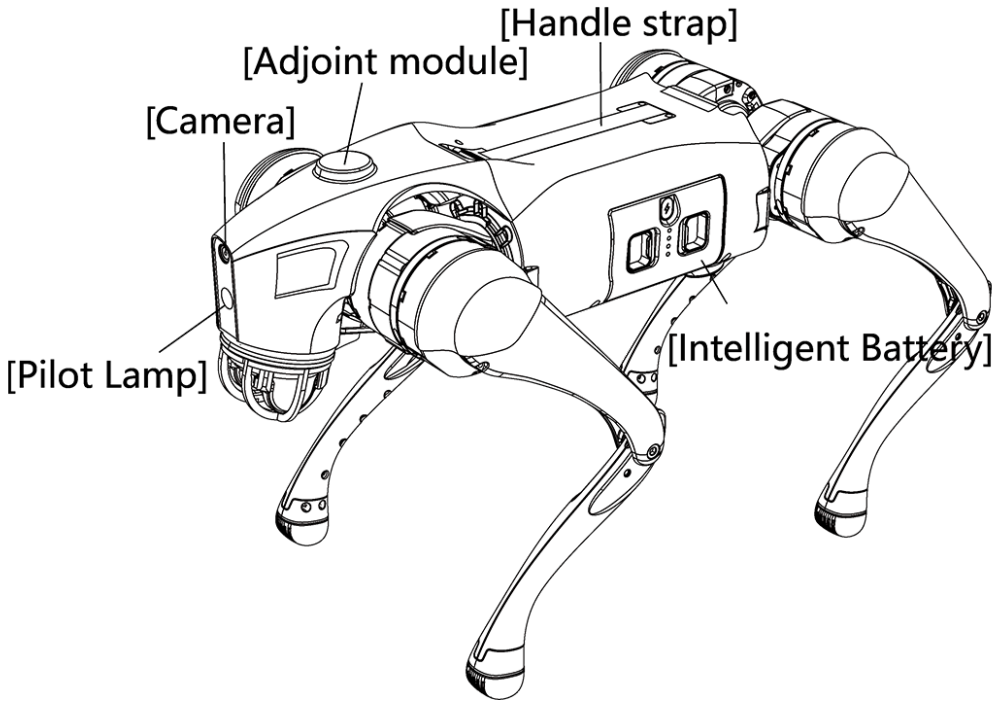
Accessory	Description	Notes	Availability	Picture
Expansion Dock (Orin Nano 8GB)	High-performance module	Up to 40 TOPs computing power	Optional	
Expansion Dock (Orin NX 16GB)	Advanced high-computing module	Up to 100 TOPs computing power	Optional	
Dual Cameras	106mm × 25mm × 41mm	Weight ≈108g, 3W power	Optional	
Depth Camera (Intel D435i)	124mm × 29mm × 26mm	1280×720@30fps, 848×480@90fps	Optional	
3D Navigation Radar (MID-360)	65mm × 65mm × 60mm	Horizontal FOV 360° / Vertical 59°	Optional	
3D Navigation Radar (Hesai XT16)	Ø100–103mm × 76mm	Horizontal 360° / Vertical 30°	Optional	
Small Servo Arm (D1 / D1-550)	Robotic manipulator arm	6 DOF + claw clamp	Optional	
Protective Bracket (Small)	Support fixture	Provides hardware protection	Optional	

Hardware Architecture


Data Transmission Module
Accompanying Module
Battery Level
Automatic Charging Control
IMU
Status
Search Light
Power Management
Fan Control
Bottom Controller
Camera
motor unit
Geophysical Radar Module
High performance CPU
Voice Module
USB
Speech Recognition
Wi-Fi / BLE
4G
Debugging Interface RJ45
Exchange Board
External Installation
Orin NX
Navigation LiDAR
Debugging Interface RJ45

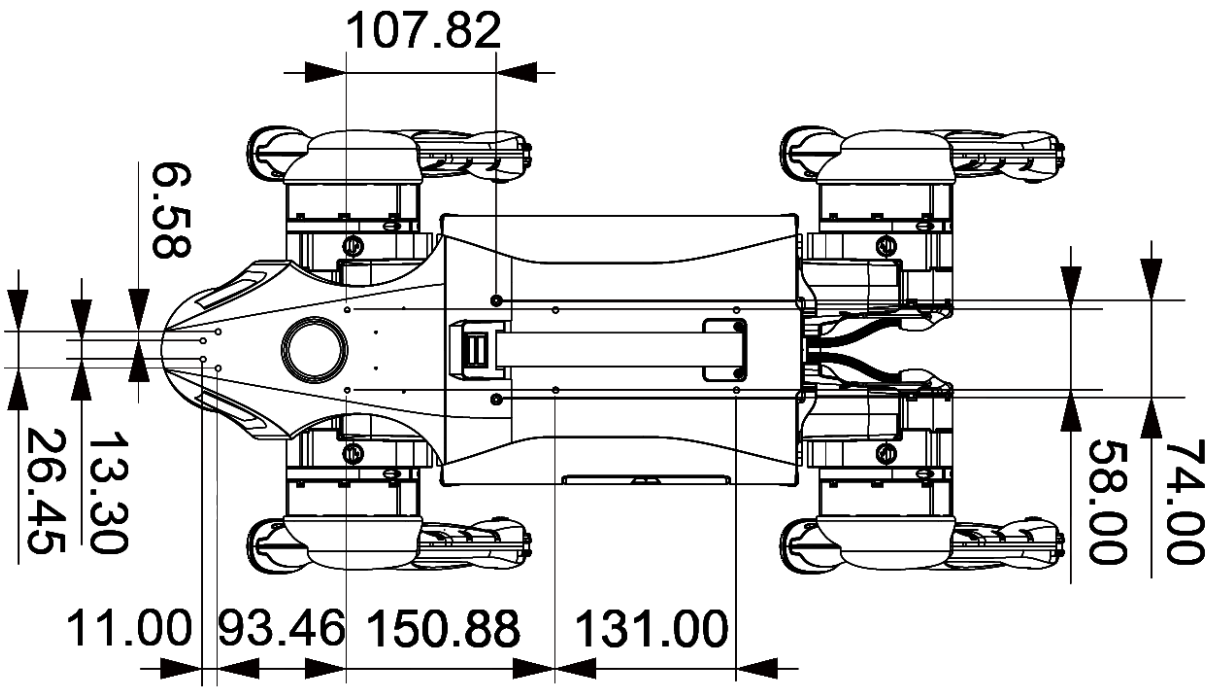
Hardware


[Robot Dog Head]
[Robot Dog Body]
[Hip]
[Thigh]
[Unitree 4D LiDAR L1]
[Knee]
[Lower Leg]
[Foot]

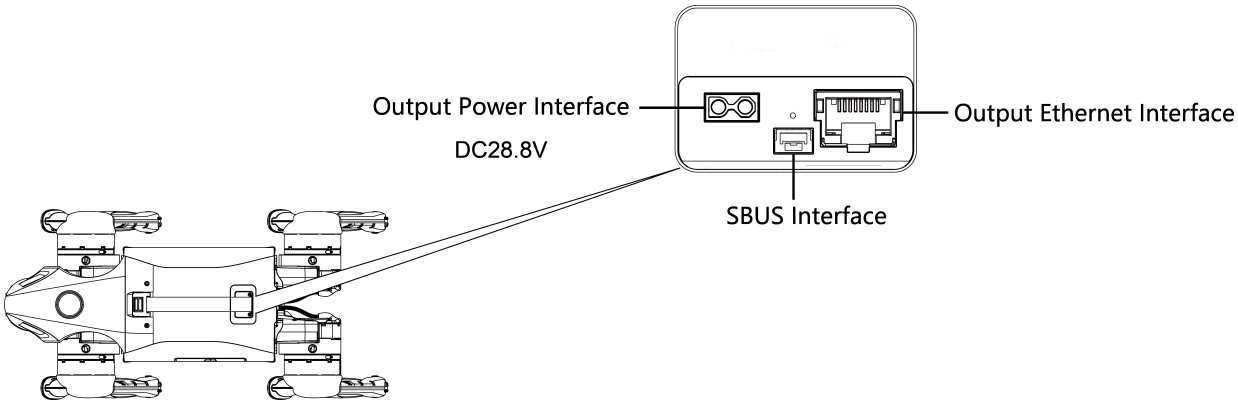


Back Load Installation Hole

> **Note**
>
> The installation holes are designed for standard M4 screws and units are in millimeters (mm).



Electrical Interface



Power Interface

DC 28.8V output Connected to Orin NX 8/16GB high computing power module BAT input.

Ethernet Interface

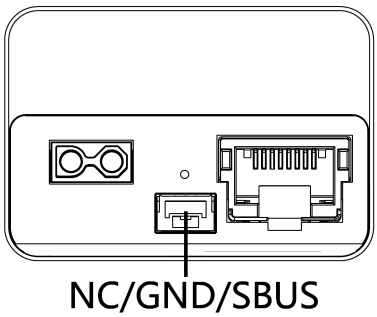
Standard RJ45 interface Connected to User PC / Orin NX 8/16GB via RJ45 Ethernet.

SBUS Interface

Used for communication connection on a universal remote control.

⚠️ *This interface does not provide power output.*

Interface definition (from left to right): `NC / GND / SBUS`



Lidar Specifications

The **Unitree 4D LiDAR-L1** is equipped with Unitree's proprietary **4D radar technology** (3D spatial positioning + 1D grayscale). It delivers **high-speed laser ranging** at up to **21,600 samples per second**, providing real-time 3D environmental data. This enables advanced **motion-based obstacle avoidance**, ensuring reliable performance across diverse and challenging terrains.

Specifications

Parameter	Details
Model	L1
Dimensions	75 mm (W) × 75 mm (D) × 65 mm (H)
Supply Voltage	12 V DC
Laser Wavelength	905 nm
Field of View	360° × 90°

Pilot Lamp Status Indicators

Color	Flashing / State	Meaning
	Flashing	Switching on
	Permanently on	Powered on, default obstacle avoidance on
	Permanently on	Obstacle avoidance closure
	Permanently on	Range mode

Color	Flashing / State	Meaning
	Permanently on	Accompanying mode on state
	Low flash rate	Motor & IMU calibration in progress
	Low flash rate	Low battery warning (crouch down in 10 min)
	Low flash rate	System abnormality, boot/hardware failure → Contact after-sales service
	Fast flash rate	Motor & IMU calibration failed
	Permanently on	Head indicator light

Notes

Light Priority Modes

The robot uses a priority-based light indication system:

Mode	Light Indicator
Companion Mode	Purple light (always on)
Endurance Mode	Yellow light (always on)
Obstacle Avoidance Off	Blue light (always on)

Camera Specifications

The onboard camera supports **dual-stream video output** at different resolutions and frame rates:

Parameter	Details
Supported Streams	720p @ 15 fps, 1080p @ 15 fps
Aperture	F2.2
Field of View	120°
Transmission	Real-time HD image transmission via app (smooth at 720p @ 15 fps in clear environments)

Joint Serial Number

A quadruped robot functions similarly to an animal, with a **trunk (body)** and four **symmetrical legs** (front and rear). The robot's four legs are grouped into **front** and **rear pairs**, each with a consistent coordinate system and joint range of motion (with slight variations between forelegs and hind legs).

Leg Numbering:

Leg	Description
0	FR – Front Right
1	FL – Front Left
2	RR – Rear Right
3	RL – Rear Left

Joint Numbering:

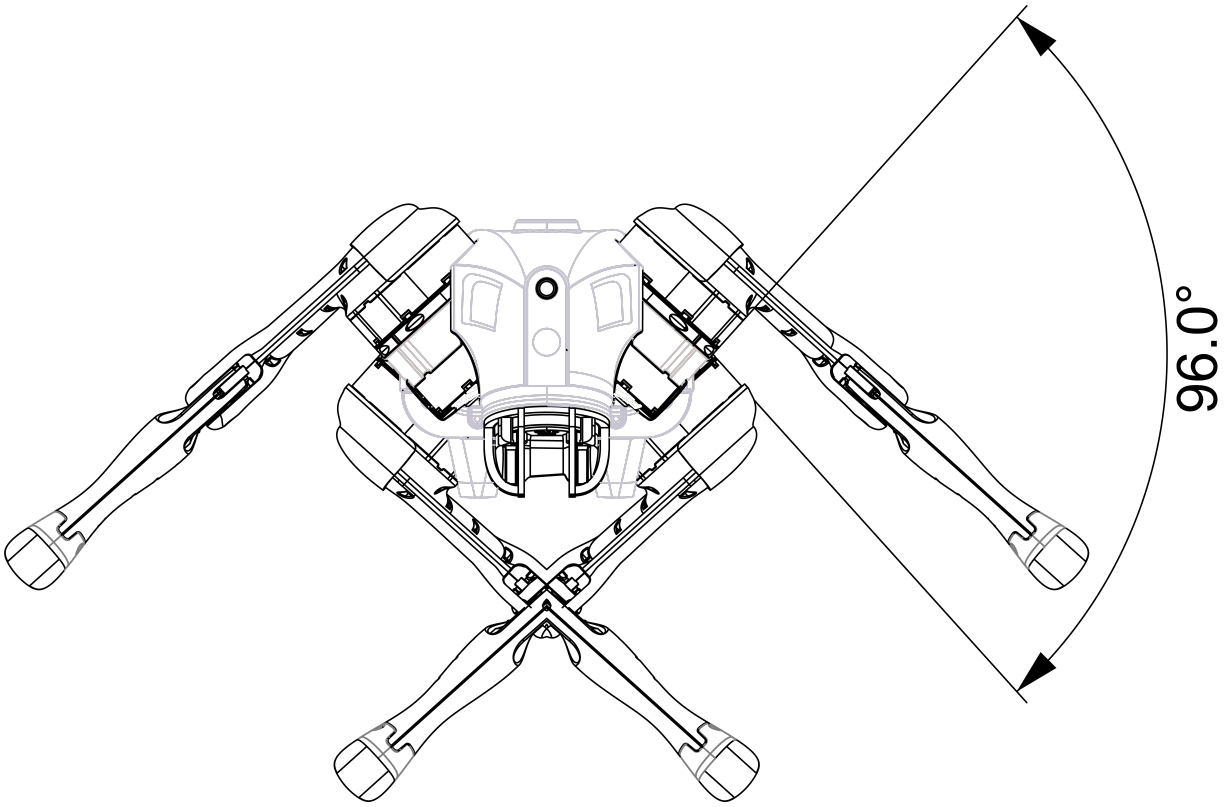
Joint	Description
0	Hip (body joint)
1	Thigh joint
2	Calf joint

Example: `FR_thigh` = Right Front Leg, Thigh Joint.

Joint Limits

Joint Motion Limits:

Joint	Range of Motion
Hip (Body Joint)	-48° ~ +48°
Thigh Joint	Vertical = 0° Forward = Positive Hind Legs: -260° ~ +30° Forelegs: -200° ~ +90°
Calf Joint	-156° ~ -48°



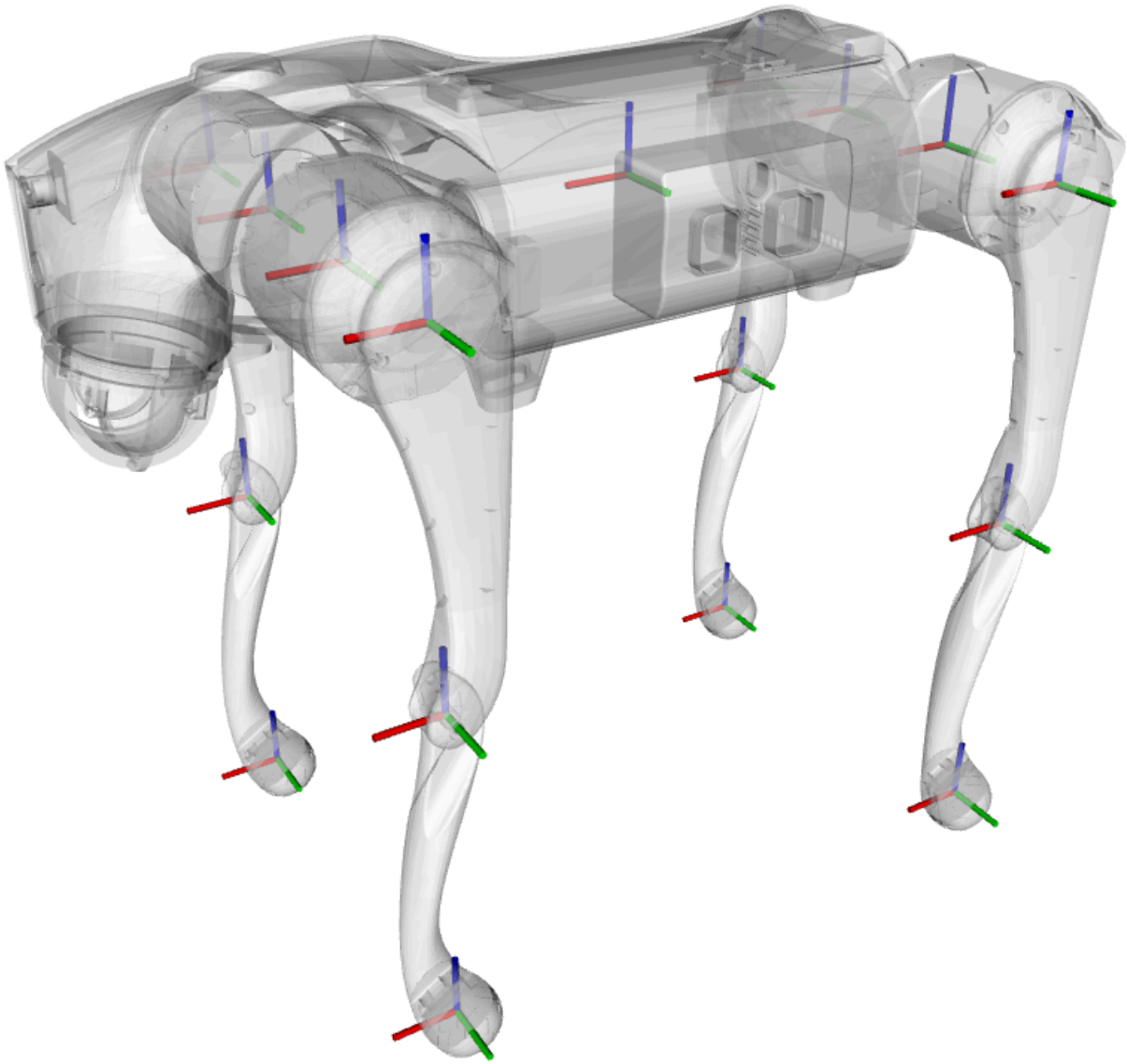
Coordinate Sys. & Joint Axes

- **Hip Joint Rotation Axis:** X-axis
- **Thigh & Calf Joint Rotation Axis:** Y-axis
- **Rotation Direction:** Positive direction follows the **right-hand rule**

When all joints are set to **zero degrees**:

- Red axis = X-axis

- Green axis = Y-axis
- Blue axis = Z-axis



Important

⚠️ Note: The calf joint cannot physically reach the zero-degree position due to its mechanical limits. The illustration only represents the **theoretical zero-point reference**.

Robot Specifications

Dimensions

Parameter	Specification
Naked machine (standing) – L × W × H	70 cm × 31 cm × 40 cm
Naked machine (lying down) – L × W × H	76 cm × 31 cm × 20 cm
Net weight (excluding battery)	~15 kg
Degrees of Freedom	12
Maximum Speed	3.7 m/s (up to 5 m/s)

Environment

Parameter	Specification
Operating Temperature	5 C ~ 35 C (optimal in fair weather)
Slope Tolerance	±40°
Maximum Step Height	16 cm
Lighting	3 W searchlight

Battery

Specification	BT2-05	BT2-06
Weight	2.5 kg	2.5 kg
Capacity	8000 mAh / 236.8 Wh	15000 mAh / 432 Wh
Rated Voltage	DC 29.6 V	DC 28.8 V
Charging Limit Voltage	DC 33.6 V	DC 33.6 V
Runtime	1 – 2 h	3 – 5 h
Charging Method	Slow charging	Fast charging / Docking
Charging Current	3.5 A	9 A
Charging Duration	~2 h	~1 h 15 min

Internal Systems

Specification	Details
Processor	8-core processor
4G Connectivity	Built-in patch SIM card
Wi-Fi	Wi-Fi 6 Dual Band Wireless (802.11x)
Bluetooth	Versions 5.2 / 4.2 / 2.1
Storage	64 GB
Output Power Supply	DC 28.8 V (battery voltage)
Connection Port	RJ45 Ethernet port

Expansion Dock

Specification	Orin Nano 8GB	Orin NX 16GB
Power Supply Voltage	16–60 V DC	16–60 V DC
Computing Power	Up to 40 TOPS	Up to 100 TOPS
Functional Interfaces	USB 3.0 Type-A ×1 USB 3.0 Type-C ×1 USB 2.0 Type-C ×1 Gigabit Ethernet (RJ45)×2 100 Mbps Ethernet (GH1.25-4PIN) ×1 M8 aviation plug ×1	USB 3.0 Type-A ×1 USB 3.0 Type-C ×1 USB 2.0 Type-C ×1 Gigabit Ethernet (RJ45)×2 100 Mbps Ethernet (GH1.25-4PIN) ×1 M8 aviation plug ×1

3D Navigation Radar

Go2 supports two types of LiDAR (MID-360 and Hesai XT16R), both compatible with SLAM navigation.

Specification	MID-360	Hesai XT16R
Dimensions	65 × 65 × 60 mm	Ø100.0/103.0 × 76 mm
Power Supply Voltage	9–27 V DC	9–36 V DC
Laser Wavelength	905 nm	905 nm
Field of View	Horizontal 360° Vertical -7° ~ +52°	Horizontal 360° Vertical -15° ~ +15°

Depth Camera

Specification	D435i
Size	124 × 29 × 26 mm
Minimum Depth Distance	0.105 m
Depth Resolution	1280 × 720 @ 30 fps 848 × 480 @ 90 fps
Field of View	86° × 57° (±3°)

Small Servo Arm

Specification	D1	D1-550
Weight	~2.37 kg	~3.15 kg
Degrees of Freedom	6 DOF + 1 claw clamp	6 DOF + 1 claw clamp
Load Capacity	500 g (ideal)	500 g (incl. clamp)
Max Arm Span	550 mm (excl. clamp) 670 mm (incl. clamp)	550 mm (excl. clamp) 670 mm (incl. clamp)
Motor Type	Servo motor	Servo motor
Working Radius	—	550 mm
Power Demand	24 V 2.5 A (max 5 A)	24 V 10 A (15–48 V)
Power	60 W	240 W
Controller	—	Integrated (4× Cortex- A55 @ 1.8 GHz)
Communication	RJ45 (Ethernet)	RJ45 (Ethernet 100 Mbps) + Type-C (debug)
Control Method	DDS subscription	DDS subscription
Control Cycle	—	10 Hz
Torque (per joint)	—	J0: 3.3 Nm J1: 3.3 Nm J2: 1.7 Nm J3: 1.7 Nm J4: 1.7 Nm J5: 1.7 Nm J6: 1.7 Nm
Joint Motion Range	J1: ±135° J2: ±90° J3: ±90° J4: ±135° J5: ±90° J6: ±135°	J0: ±135° J1: ±90° J2: ±90° J3: ±135° J4: ±90° J5: ±135° Claw: 0–65 mm

Remote Control with Screen

Specification	Details
Display	5.5″ 1080p HD LCD touchscreen
System	Android 9, 2 GB RAM, 16 GB storage
Size (antenna retracted)	189 × 138 × 41 mm
Weight	850 g
Battery	10,200 mAh, 7.4 V, 2S Li-ion
Charging Protocol	PD 20 W (fast charging)
Charging Time	~5 h (20 W fast charging)
Runtime	Up to 13 h
Functional Interfaces	HDMI (video out, industry version only) USB-A (external device / file transfer) Type-C (charging & transfer) Firmware upgrade port (GH1.25 4PIN) SIM card slot TF card slot Tripod mount (1/4″ threaded hole)

Dual Cameras

Specification	Details
Size	106 × 25 × 41 mm
Weight	108 g
Power Consumption	3 W
Searchlight Power	8 W
Luminous Flux	120–140 lm @ 1 W
Searchlight Type	Transparent

Dual Cameras + Three-in-One Module

Specification	Details
Size	270 × 143 × 133 mm
Weight	690 g ± 10 g
Power Supply Voltage	16–60 V DC
Brightness	3000 lm
Light Output Angle	15°
Power	Lighting: 30 W Speaker: 30 W Red/Blue: 10 W
Effective Sound Range	500 m
Maximum Sound Range	800 m
Maximum Sound Pressure	130 dB
Audio Formats	mp3 / wav / m4a / flac / aac
Control Method	Remote app
Functions	Real-time shouting Audio recording & playback Text-to-speech Voice + lighting control Flash / constant light Brightness adjustment Red/Blue flash & mode switching

‹ GO2 Quadruped Tutorials

Packaging Description ›

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